DUPLEX FILTER CONTROL DESCRIPTION

GENERAL OPERATION- the system is designed so that A and B filters run in parallel with both units normally in service at the same time. There are three conditions that can trigger a backwash sequence. All triggers are followed by a time of day delay. The backwash trigger method is selectable through the HMI. A backwash sequence is defined as backwash and a fast rinse. After a backwash sequence the unit will return to service. If a unit is in backwash or fast rinse the other unit goes offline. Only one unit can be in a backwash sequence at a time. The three types of backwash triggers are:

1. Differential Pressure (DP) Trigger - There are four pressure transmitters (located on the Inlet and Outlet of each vessel) which are used to determine the differential pressure for each filter (A and B). There is an adjustable delay that can be entered on the HMI to avoid a quick DP spike from triggering a backwash. After the system's DP has risen above the setpoint for the duration of the adjustable delay, there will be an alarm and the units will delay until the time of day selected for filter A, and then the unit will backwash sequentially - A followed by B (five minutes after). Each unit has it's own time of day delay, but in the case of a DP triggered event, they will only heed the time of day for vessel A. An alarm will be displayed notifying the operator that a high DP condition has triggered a backwash.

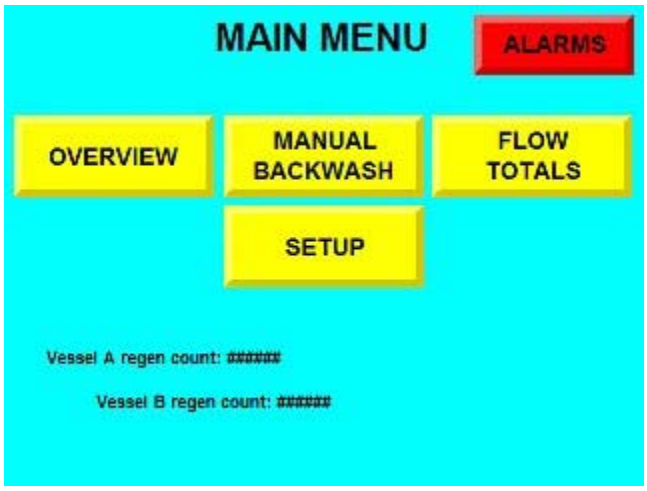
2. Elapsed Run Time (Hours Left) - each unit has an entry for elapsed service run time. This value is entered in days. The time left in days will be displayed on this screen if the mode chosen to start backwash is elapsed time. Once this elapsed run time in service has been reached and the time of day for each vessel (A and B) arrives, that unit will go into backwash.

3. Service Gallons (Gallons Remaining) - each unit has an entry for batch gallons. The unit will count service gallons for each vessel (A and B), and when it reaches the setpoint and the time of day for the same vessel, it will go into backwash.

Following is a detailed explanation of all the screens:

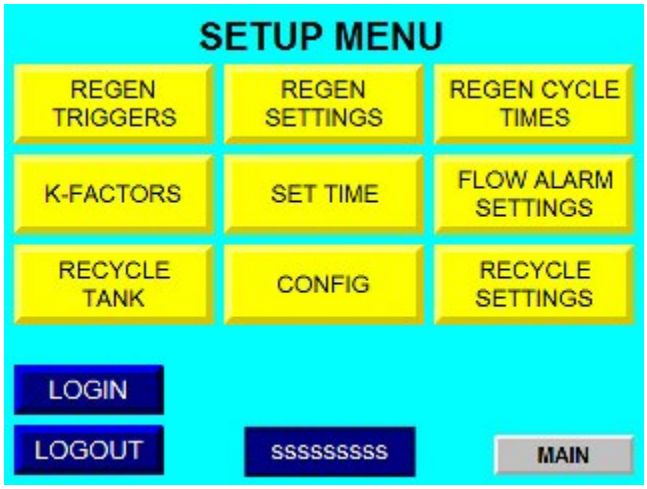
MAIN MENU

This screen can be used to navigate to any of the listed screens.



SETUP MENU

The setup menu screen allows the operator to navigate to any of the screens listed. The LOGIN / LOGOUT function enables access to K-FACTORS and CONFIG, which are only required for advanced maintenance of the system.



OVERVIEW SCREEN

The overview screen displays a graphic layout of the filter system.

FLOW - Inlet GPM is displayed on tanks A and B, as well as a system overall GPM on the inlet header. GPM for tanks C and D are indicated in the upper left corner of the screen.

VESSEL STAGE INDICATORS - below each tank is an indicator showing its current stage. It can be any of the following:
Service - the unit is producing product water
Backwash - 1st cleaning cycle
Fast rinse - 2nd step in cleaning cycle
Offline - unit is forced offline and not available for service.

VESSEL MODE INDICATORS - below each tank is an indicator showing its current operating mode. It can be either manual or automatic.

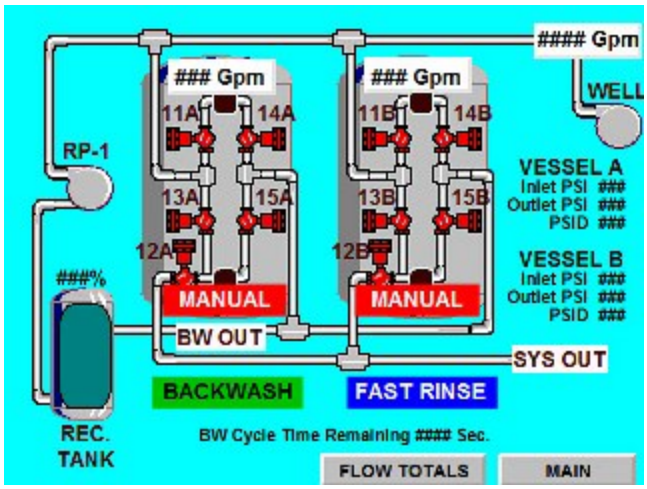
CYCLE TIME - if a unit is in a backwash or fast rinse step the cycle time will count down here in seconds.

PUMP RUN INDICATORS- The well and recycle pumps each have an indicator on the screen. They will display green when running or red when not.

TANK LEVEL INDICATOR – The tank will fill to indicate the level of the Recycle Tank.

AUTOMATIC VALVE POSITION INDICATORS - each automatic valve will be green when commanded open and red when commanded close.

NAVIGATION - Aside from the buttons taking you to the FLOW TOTALS screen and the MAIN MENU, the vessels and tanks are all clickable and take you to their respective control or setup screens.


Gpm
Gpm
Gpm
WELL
11A
14A
11B
14B
RP-1
VESSEL A
Inlet PSI ###
Outlet PSI ###
PSID ###
13A
15A
13B
15B
12A
12B
###%
VESSEL B
Inlet PSI ###
Outlet PSI ###
PSID ###
MANUAL
MANUAL
BW OUT
SYS OUT
BACKWASH
FAST RINSE
REC.
TANK
BW Cycle Time Remaining ### Sec.
FLOW TOTALS
MAIN

VESSEL X MANUAL CONTROL

DAYS UNTIL BW - If the days-based regen trigger is enabled, the days remaining until the next cycle will be displayed in the lower left corner of the screen.

CAP REMAINING - If the volume-based regen trigger is enabled, the gallons remaining until the next cycle will be displayed in the lower left corner of the screen.

REGEN CYCLE TIME REMAINING - If the vessel is in backwash or fast rinse, the time remaining until the next stage will be displayed in the lower left corner of the screen.

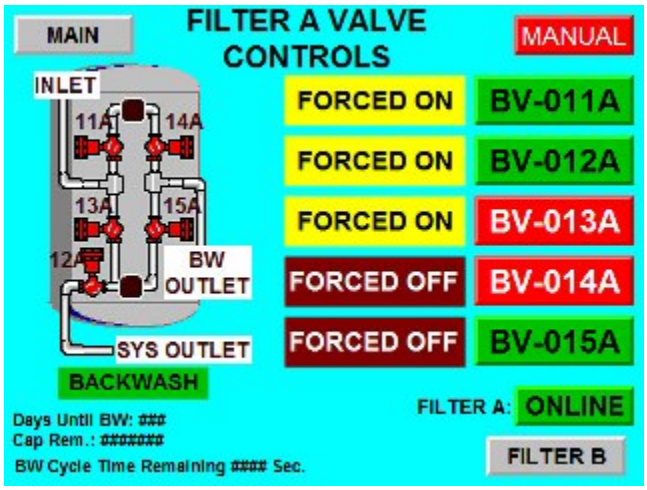
MANUAL / AUTO BUTTON - Toggle the valves on this vessel between manual and automatic control.

ONLINE / OFFLINE BUTTON - Enable or completely disable the valves on this vessel. Offline closes all valves and prevents them from being actuated.

VALVE STATUS INDICATORS - These show the current status of the valve (labled to the right).

VALVE LABELS / ACTUATOR BUTTONS - In manual control mode, clicking on the valve lables will toggle the respective valve between open and closed positions.

NAVIGATION - Aside from the buttons taking you to the MAIN MENU screen and FILTER X, the vessel is clickable and takes you back to the SYSTEM OVERVIEW screen.



MANUAL BACKWASH

The operator can use this screen to manually start a backwash sequence, advance the stage of a cycle, or abort a cycle completely. The timing of each step is still automatic, unless advanced manually by the operator. The status of each unit is shown near the top for reference. Note that backwash controls only function when the respective vessel is in auto control mode.

BACKWASH START BUTTON- Press the backwash start button to start the desired unit into a backwash. This button can only be used to start. It can't be used to advance the unit once it is started.

BW ABORT BUTTON- use this button to end the backwash sequence and return the unit back to service.

ADVANCE BUTTON- Use this button once a backwash cycle has begun to advance the unit from backwash to fast rinse, and from to fast rinse back to service.

MANUAL / AUTO BUTTON - Toggle the valves on this vessel between manual and automatic control.

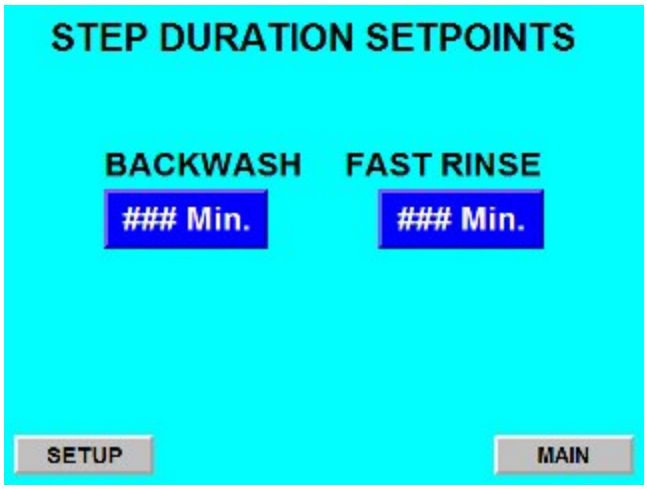
NAVIGATION - A button in the lower right corner of the screen will take you back to the MAIN MENU screen.



CYLE TIME ENTRIES

Use this screen to adjust the times for the backwash and fast rinse steps. All times are in minutes.

NAVIGATION - A button in the lower right corner of the screen will take you back to the MAIN MENU screen. On the lower, left, a SETUP button takes you to the SETUP MENU.



REGEN TRIGGER SELECT

This screen can be used to select from the regen triggers which will initiate a cycle, and to configure the relevant setpoints for all triggers.

TRIGGER - Choose either Days or Gallons (Time or Volume) to trigger a regen cycle. PSID (Differential Pressure) can be enabled or disabled independently of whether Time or Volume is being used.

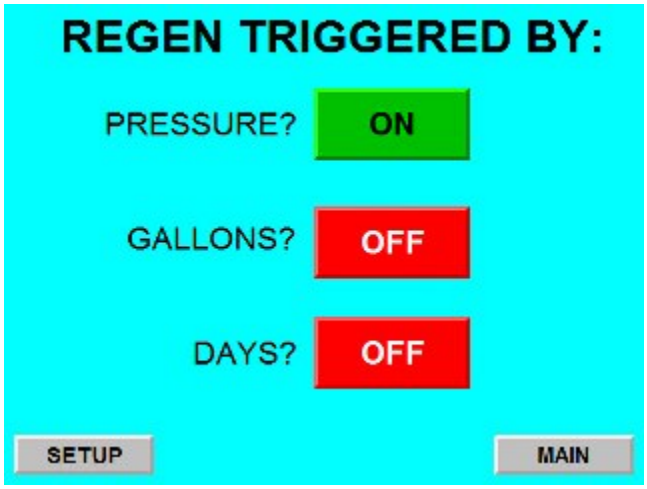
DELAY - Enter how many seconds of continuously high differential pressure are required to trigger an alarm condition and / or initiate a cycle.

PRESSURE – Press this to enable DP triggering. Green indicates the trigger is enabled; red means disabled.

GALLONS – Press this to enable volume triggering. Green indicates the trigger is enabled; red means disabled.

DAYS – Press this to enable time triggering. Green indicates the trigger is enabled; red means disabled.

NAVIGATION - A button in the lower right corner of the screen will take you back to the MAIN MENU screen. A button in the lower left corner of the screen will take you back to the SETUP MENU screen.



REGEN TRIGGER SETTINGS

This screen can be used to select from the regen triggers which will initiate a cycle, and to configure the relevant setpoints for all triggers.

TRIGGER - Choose either Days or Gallons (Time or Volume) to trigger a regen cycle. PSID (Differential Pressure) can be enabled or disabled independently of whether Time or Volume is being used.

DELAY - Enter how many seconds of continuously high differential pressure are required to trigger an alarm condition and / or initiate a cycle.

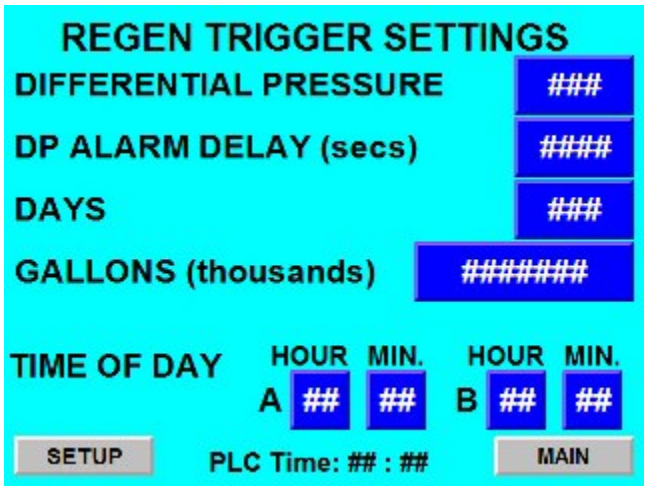
DAYS - The days of service for each vessel before a cycle is triggered can be entered here.

GALLONS - The gallons through each vessel before a cycle is triggered can be entered here. Note: this entry is made in thousands of gallons.

TIME OF DAY - Enter an hour (0-23) and a minute for each vessel which will specifi what time of day regen cycles will occur once triggered. Note: times for vessels A and B cannot be set within 30 minutes of one another. The system will automatically correct the time for vessel B if the times are too close.

PLC TIME - This is shown here as a convenience.

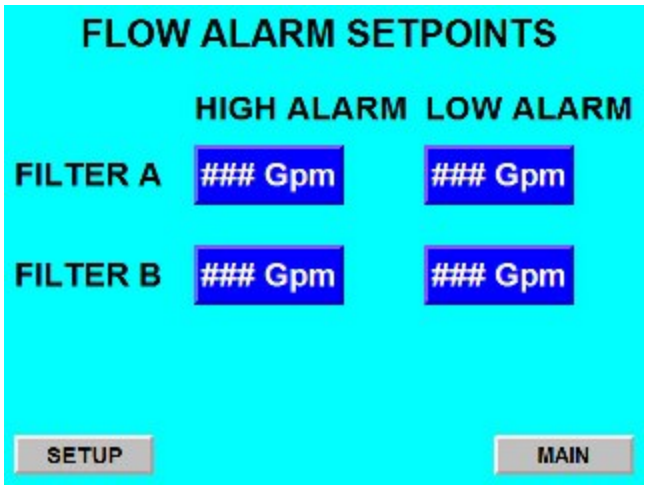
NAVIGATION - A button in the lower right corner of the screen will take you back to the MAIN MENU screen. A button in the lower left corner of the screen will take you back to the SETUP MENU screen.



SERVICE FLOW ALARMS

This screen can be used to set the high and low alarms for each unit. There is a short delay on this alarm to avoid false alarms from surging.

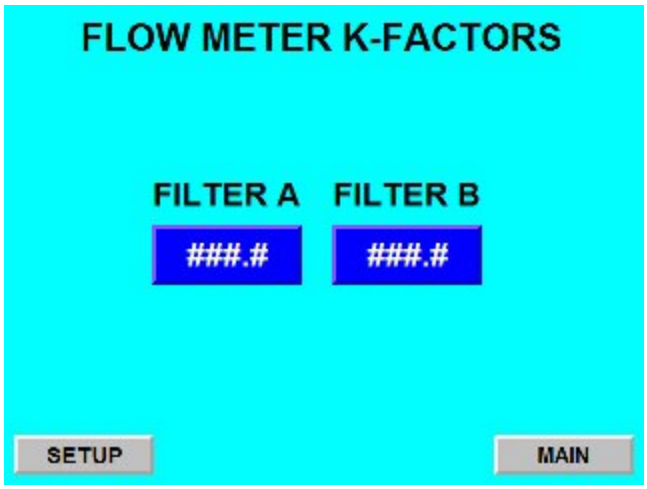
NAVIGATION - A button in the lower right corner of the screen will take you back to the MAIN MENU screen. A button in the lower left corner of the screen will take you back to the SETUP MENU screen.



K FACTORS

Use this screen to enter in the K FACTOR for each flow sensor. This is the pulses per gallon factor that is determined by the flow sensor and type of fitting used.

NAVIGATION - A button in the lower right corner of the screen will take you back to the MAIN MENU screen. A button in the lower left corner of the screen will take you back to the SETUP MENU screen.



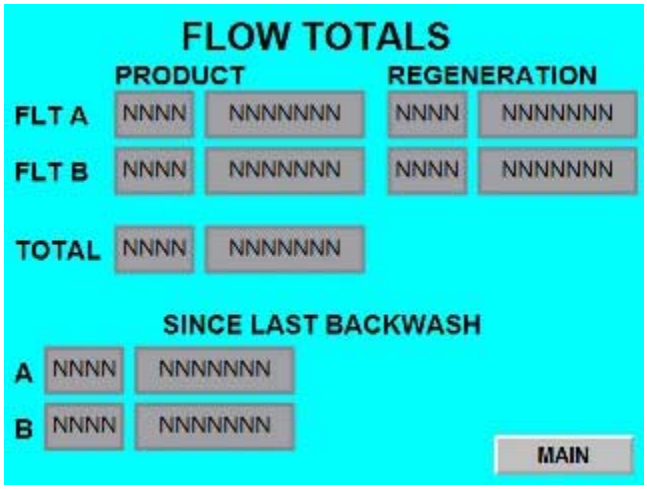
FLOW TOTALS

This screen displays the flow totals for each unit- product, backwash and since last backwash.

"Since last Backwash" is reset automatically every time the unit ends backwash and goes back into service. All other flow totals are not re-settable unless you contact the factory.

The total is calculated by adding the flows for vessels A and B.

NAVIGATION - A button in the lower right corner of the screen will take you back to the MAIN MENU screen.



RECYCLE SYSTEM CONTROL SCREEN-

Use this screen to adjust the set points of the recycle tank level transmitter and to control the Recycle Pump. This screen also has entries for the following presets:

HAND - When the Recycle Pump is Online and in Manual control mode, use this to start or stop the device.

ON/OFF - Use this to toggle the Recycle Pump between Online and Offline. Offline means it cannot be started at all.

AUTO - When the Recycle Pump is Online use this to toggle the Recycle Pump between automatic and manual control.

STATUS INDICATOR - This indicator will display the current status of the Recycle Pump.

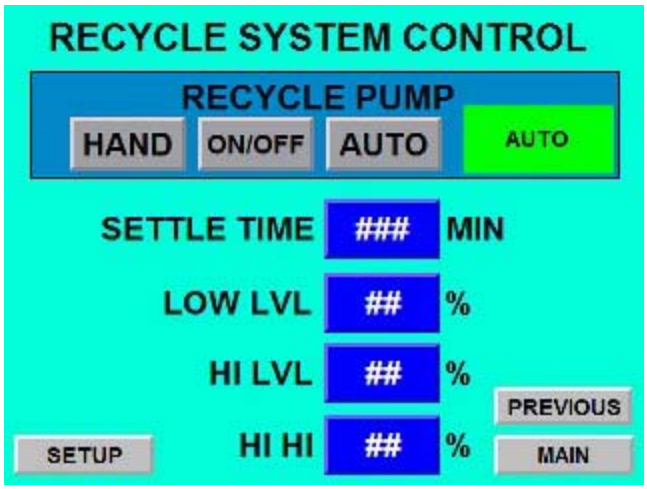
SETTLE TIME - Enter how many minutes the pump should wait after a backwash before it starts pumping the wastewater back into the inlet stream.

LOW LEVEL - Once the recycle tank level falls to this level, the recycle pump will stop and a regen cycle will be premitted to initiate.

HIGH LEVEL - If the recycle tank level reaches this setpoint, a backwash will not be allowed to initiate automatically.

HIGH HIGH LEVEL - If the recycle tank level reaches this setpoint, an alarm will be triggered.

NAVIGATION - A button in the lower right corner of the screen will take you back to the MAIN MENU screen. A button in the lower left corner of the screen will take you back to the CONFIG MENU screen. The PREVIOUS button will take you back to the last screen visited.



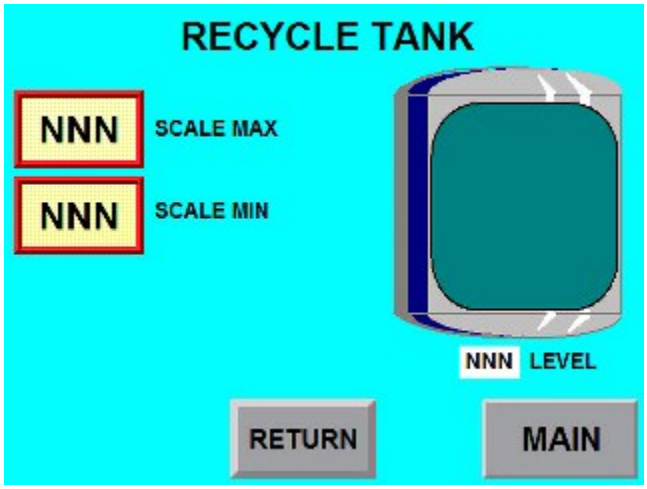
RECYCLE TANK SETUP SCREEN-

Use this screen to adjust the scaling of the recycle tank level transmitter.

SCALE MAX – Use this to set the high range of the level transmitter.

SCALE MIN - Use this to set the low range of the level transmitter.

NAVIGATION - A button in the lower right corner of the screen will take you back to the MAIN MENU screen. The RETURN button will take you back to the last screen visited.



CONFIGURATION SCREEN

This screen is used to exit the HMI program. Exiting the HMI program will NOT impede system operation. Factory credentials are required to access this screen as it is typically only required for advanced maintenance and configuration.

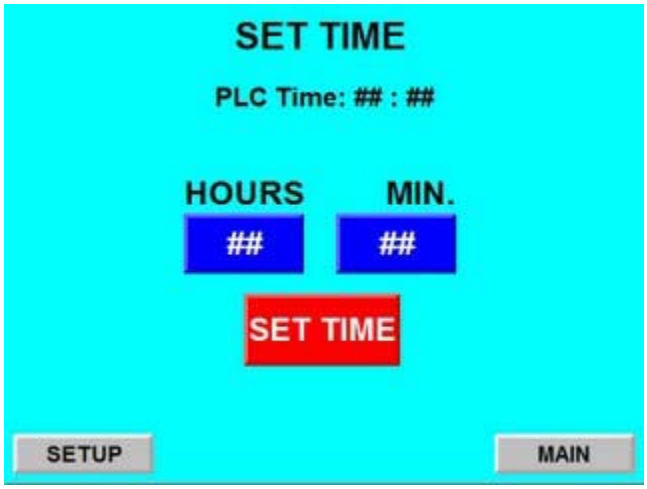
NAVIGATION - A button in the lower right corner of the screen will take you back to the MAIN MENU screen.



SET TIME

Use this screen to adjust the realtime clock. The top row shows the current time. The middle area has two buttons for minutes and hours. Use these two buttons to enter in the correct time. Use the red "set time" button to send the new value to the controller. Note: this time is important for the regen cycle time of day (time of day to start a backwash), but is NOT related to the time of day displayed with an alarm. That time is set in the screen by exiting the program.

NAVIGATION - A button in the lower right corner of the screen will take you back to the MAIN MENU screen. A button in the lower left corner of the screen will take you back to the SETUP MENU screen.



ALARM HISTORY

ALARM HISTORY- this screen records the alarms.

Below is a list of alarms that can occur:

1. SYSTEM HIGH DP - This occurs when the system has had high DP. This alarm occurs regardless if the unit has its DP start enabled.

2. HIGH FLOW - This occurs if the flow rate for some vessel goes above setpoint.

3. LOW FLOW - This occurs if the flow rate for some vessel goes below setpoint.

4. CIRCUIT BREAKER TRIPPED - This occurs when a circuit breaker trips or is opened.

Any of the above alarms will first cause an alarm banner to appear. Use the button on the banner to close it and then press the alarm silence button on any of the screens to silence the horn.

NAVIGATION - The MAIN button in the lower left corner of the screen will take you back to the MAIN MENU screen.

